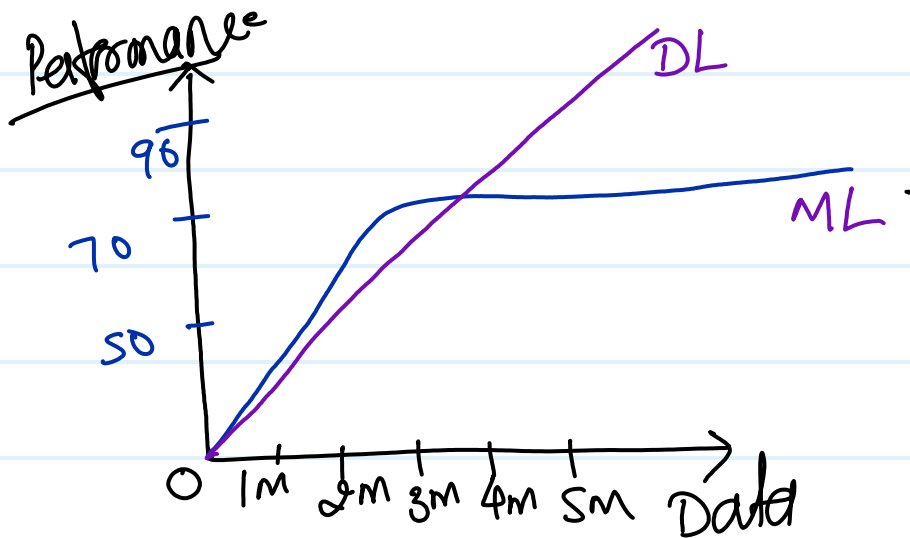


# Deep learning

# To mimic human brain



1Y = 10GB

1Y = 1TB

2008 → smartphone → Data ↑ - 100GB

2010 → Data ↑ → 2000GB

2012 → ✓ Big Data (Data Engg) → 10000GB

2014-15 → Data science (ML) ✓

2018-20 → Data science (ML + DL)

GPU

Data (information)

numba, text, image, video,  
Text, voice

ML

2D Databse

Reg/class/clust.

CPU

DL

multi D data

KV / NLP ✓

✓ GPU

2022 → AI Copilot

Nov23 → chatGpt → GPU

1 Tr. \$

→

Apple

2020-2019

800 Bill

900 Bill

100 Bill

→

Nvidia ✓

2024 → 1.7 Tri.

2026 → 4.8 Tri. ✓

India GDP → \$ 5 Tri. ✓

## Deep Learning

- 1) ANN (Artificial Neural Network)
- ② CNN (Convolutional Neural Network)
- ③ RNN (Recurrent Neural Network)

ANN → ML

Architecture      Reg. / Class. / Clustering.

CNN

visual data

[ANN + CNN] → ✓ Image classification  
 object detection  
 Image segmentation  
 Pose detection

ex:- (self driving)

RNN

⇒ language / Text / voice Recording

✓ chatbot / Alexa / Siri / Google voice  
 chatGPT / Gemini

# computer vision (cv) ← CNN

# natural Language processing (NLP)  
 ← RNN

ANN

CNN

RNN

vanilla  
architecture

Lenet

Alexnet

VGG

Inception

Res net

R-CNN

Fast R-CNN

faster R-CNN

LSTM

GRU

encoder-decoder

BERT

Attention mech.

transformer

- 1) Neuron
- 2) Perceptron
- 3) Input layer
- 4) Hidden layer
- 5) output layer
- 6) weight
- 7) bias
- 8) Activation function
- 9) Loss function
- 10) optimizer
- 11) Forward propagation
- 12) Back propagation
- 13) weight initializer
- 14) epoch
- 15) Batch size
- 16) iteration
- 17) Dropout
- 18) Base equation

$$\checkmark z = wx + b \checkmark$$

$$\checkmark y = mx + c$$

